

Data Mining Methodologies and Systems

- ❑ Structuring Unstructured Data for Knowledge Mining: A Data-Driven Approach
- ❑ Data Augmentation
- ❑ From Correlation to Causality
- ❑ Network as a Context
- ❑ Auto-ML: Methods and Systems 

Auto-ML: Methods and Systems

□ Why do we need Auto-ML

- Different data mining and machine learning models have been widely used
- Different tasks have different settings
- Building and training appropriate mining models for each task could be time consuming
- Domain experts might not be familiar with the mining models

□ Auto-ML classification

- Automated Feature Engineering (Auto-FE)
 - Automatically detects the most representative and informative features
- Automated Model and Hyperparameter Tuning (Auto-MHT)
 - Automatically builds machine learning models and tunes the hyper-parameters

Auto-ML: Methods and Systems

- ❑ Key challenges for Auto-ML
 - ❑ The lack of authoritative benchmarks for Auto-FE and Auto-MHT
 - ❑ The efficiency of Auto-ML
 - ❑ How to incorporate human knowledge or experiences into the AutoML to find suitable sub-space of the entire search space
 - ❑ The interpretability of Auto-ML
 - ❑ The scope of applications
- ❑ Taxonomy of Auto-ML

